PROJECT PORTFOLIO · ZEYNEP

Daytime Work Schedule

A program that builds fair, rule-based monthly operating-room assignments for the anesthesiology department of a public teaching hospital in Kayseri, Türkiye.

27	25	~525	0
anesthesiologists scheduled each month	operating rooms, clinics and endoscopy units	assignments per month	rule violations in test runs

All names in this portfolio are fictional (personal data protection).

October 2026

THE NEED

A monthly puzzle solved by hand

Every month one specialist assigned about 27 anesthesiologists to 25 work areas for every working day — by hand.

- Each area carries a different **workload score** (200 to 5,000 points), and doctors expect their monthly totals to be **equal**.
- When someone falls ill or a room closes mid-month, everything has to be reworked — and reported to the hospital administration.
- The schedule must respect many interlocking rules: the day off after a night shift (different for Friday and weekend shifts), leave, 1–2 clinic and endoscopy days per person in different units, rooms a doctor is exempt from, at most two days in the same room, which rooms may be combined when doctors are short…

Goal: a program that never breaks a rule, makes the monthly totals as equal as possible, and fits the department's real workflow.

ŞEHİR EĞİTİM VE ARAŞTIRMA HASTANESİ ANESTEZİYOLOJİ VE REANİMASYON KLİNİĞİ 2026 YILI KASIM AYI DOKTOR ÇALIŞMA LİSTESİ

TARİH	POLK-1	1-5-7 ODA	2 ODA	3-4 ODA	6-8 ODA	9 ODA	10 ODA	11-12.ODA	13 ODA	14 ODA	15-18 ODA	16-17-23
01/11/2026												
02/11/2026	KEREM ASLAN	prof.dr. A. KARA	EMRE KILIÇ	TOLGA AVCI	NAZLI KOÇ	KEMAL ARSLAN	SELİN AYDEMİR	OZAN ÇELİK	MURAT EKİNCİ	ASLI ÖZDEMİR	ELİF YILDIZ	doç.dr. B. TUNA
03/11/2026	ECE KARATAŞ	prof.dr. A. KARA	SELİN AYDEMİR	MURAT EKİNCİ	ASLI ÖZDEMİR	CANAN GÜLER	ONUR ŞAHİN	KEMAL ARSLAN	HANDE ATEŞ	İREM AKSOY	VOLKAN IŞIK	doç.dr. B. TUNA
04/11/2026	HAKAN DEMİR	prof.dr. A. KARA	ECE KARATAŞ	MELİS KARACA	TUBA KURT	İREM AKSOY	TOLGA AVCI	MURAT EKİNCİ	BARIŞ DUMAN	KEREM ASLAN	SİBEL AKIN	doç.dr. B. TUNA
05/11/2026	ONUR ŞAHİN	prof.dr. A. KARA	NAZLI KOÇ	GİZEM ÇINAR	ELİF YILDIZ	SELİN AYDEMİR	SİBEL AKIN	EMRE KILIÇ	EMRE KILIÇ	SİBEL AKIN	MURAT EKİNCİ	doç.dr. B. TUNA
06/11/2026	NAZLI KOÇ	prof.dr. A. KARA	İREM AKSOY	BARIŞ DUMAN	GİZEM ÇINAR	SİBEL AKIN	ECE KARATAŞ	CANAN GÜLER	CANAN GÜLER	KEMAL ARSLAN	MELİS KARACA	doç.dr. B. TUNA
07/11/2026												
08/11/2026												
09/11/2026	UĞUR TAŞ	prof.dr. A. KARA	CEM YAVUZ	KEMAL ARSLAN	ELİF YILDIZ	TOLGA AVCI	TUBA KURT	HAKAN DEMİR	ECE KARATAŞ	BARIŞ DUMAN	İREM AKSOY	doç.dr. B. TUNA
10/11/2026	CAN ÖZTÜRK	prof.dr. A. KARA	ELİF YILDIZ	ASLI ÖZDEMİR	SELİN AYDEMİR	ECE KARATAŞ	NAZLI KOÇ	ONUR ŞAHİN	İREM AKSOY	MURAT EKİNCİ	HANDE ATEŞ	doç.dr. B. TUNA
11/11/2026	MELİS KARACA	prof.dr. A. KARA	VOLKAN IŞIK	CANAN GÜLER	İREM AKSOY	HANDE ATEŞ	OZAN ÇELİK	KEREM ASLAN	NAZLI KOÇ	CAN ÖZTÜRK	SELİN AYDEMİR	doç.dr. B. TUNA
12/11/2026	SİBEL AKIN	prof.dr. A. KARA	KEMAL ARSLAN	NAZLI KOÇ	ONUR ŞAHİN	HAKAN DEMİR	KEREM ASLAN	GİZEM ÇINAR	UĞUR TAŞ	BURAK TEKİN	EMRE KILIÇ	doç.dr. B. TUNA
13/11/2026	CANAN GÜLER	prof.dr. A. KARA	SİBEL AKIN	TUBA KURT	EMRE KILIÇ	MURAT EKİNCİ	HANDE ATEŞ	TOLGA AVCI	BARIŞ DUMAN	NAZLI KOÇ	BURAK TEKİN	doç.dr. B. TUNA
14/11/2026												
15/11/2026												
16/11/2026	KEMAL ARSLAN	prof.dr. A. KARA	CAN ÖZTÜRK	UĞUR TAŞ	OZAN ÇELİK	EMRE KILIÇ	CEM YAVUZ	İREM AKSOY	OZAN ÇELİK	GİZEM ÇINAR	GİZEM ÇINAR	doç.dr. B. TUNA

The hospital's official monthly list (Excel, in the administration's own template; first columns shown) — now produced automatically.

HOW IT WORKS

Built around the department's real monthly workflow

1 · Calendar & night-shift roster
holidays, room availability, the four duty columns

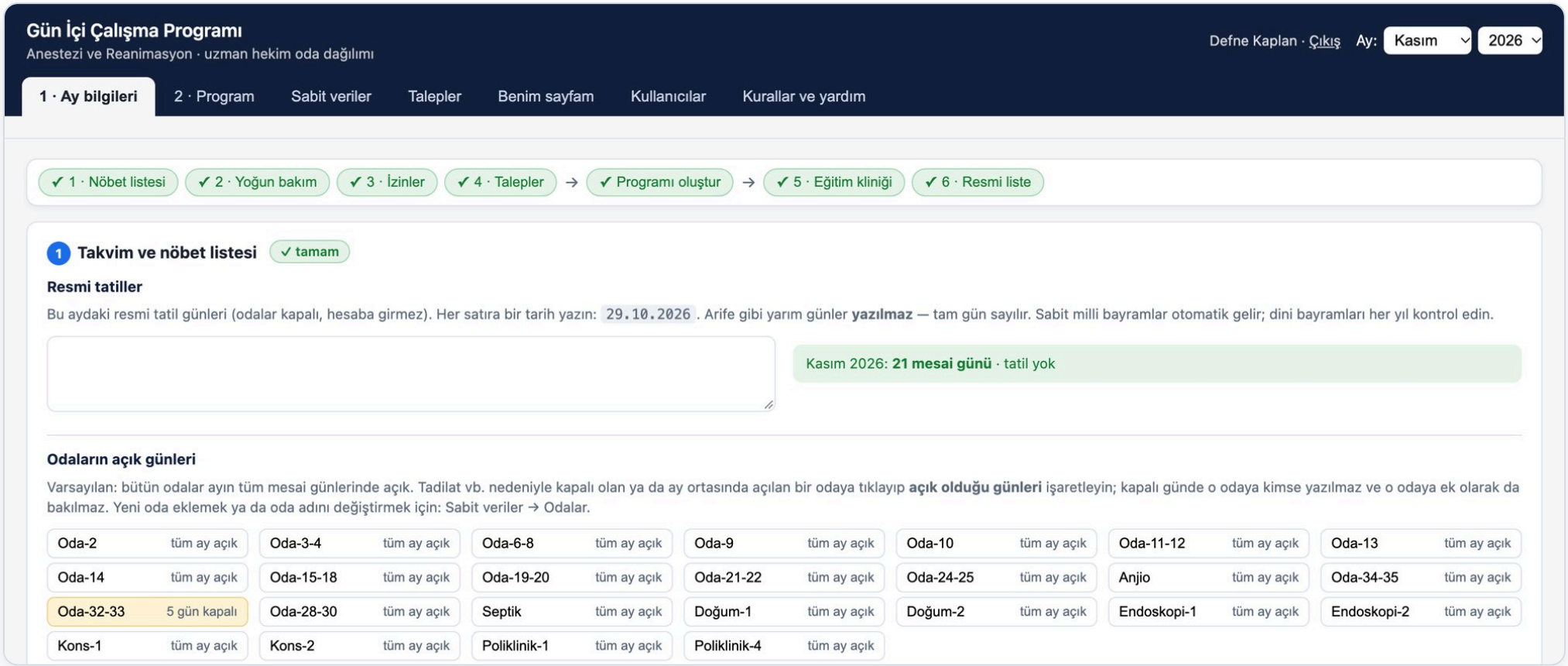
2 · Intensive-care staff
who is outside the daytime schedule this month

3 · Leave
annual, congress, sick leave — entered by doctors themselves

4 · Personal requests
preferred clinic days, rooms to avoid on given days

→ Generate & review
the algorithm builds the month in seconds

5–6 · Training clinic & official list
merged into the administration's Excel template



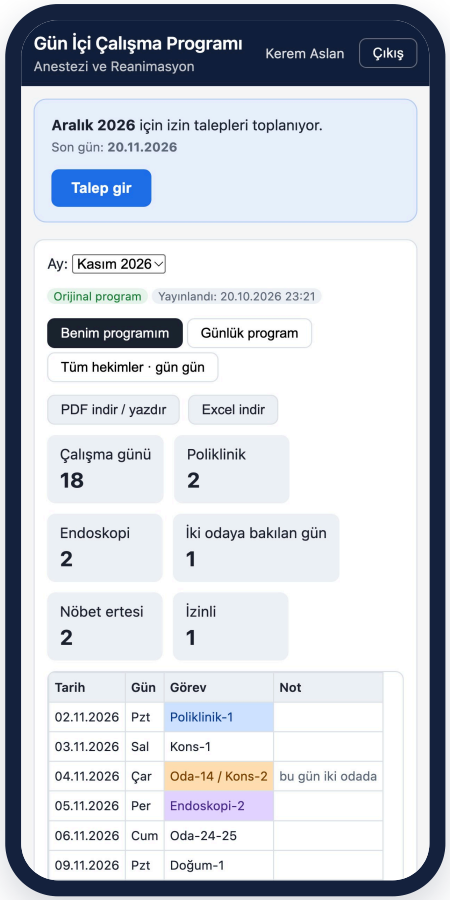
Step-by-step input screen: a status bar shows which steps are complete (✓) and which still need attention.

LEAVE REQUESTS

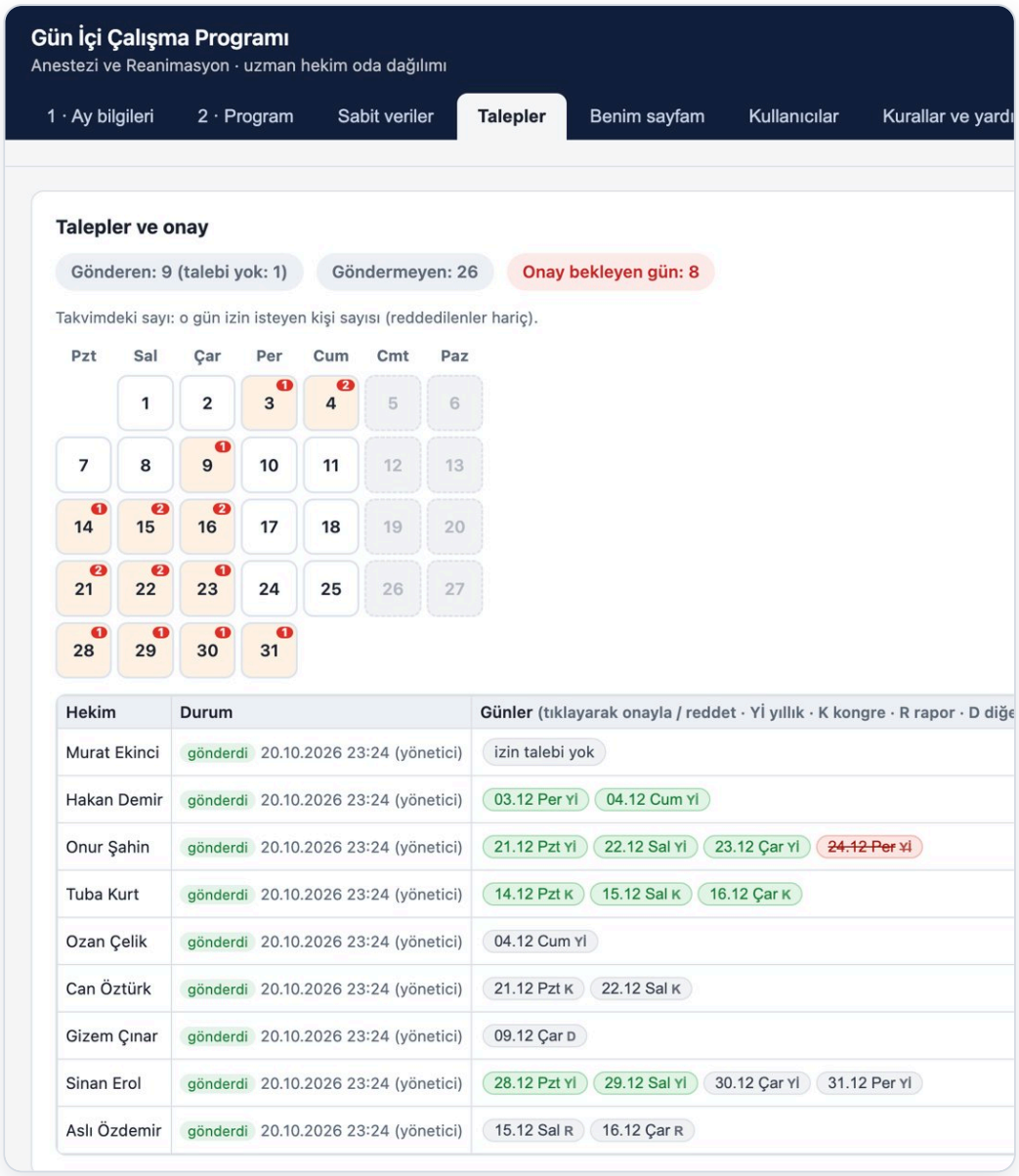
Doctors enter their own leave from their phones



Choose the leave type, tap the days, send. Nothing reaches the scheduler until "Send" is pressed — the page warns before leaving.



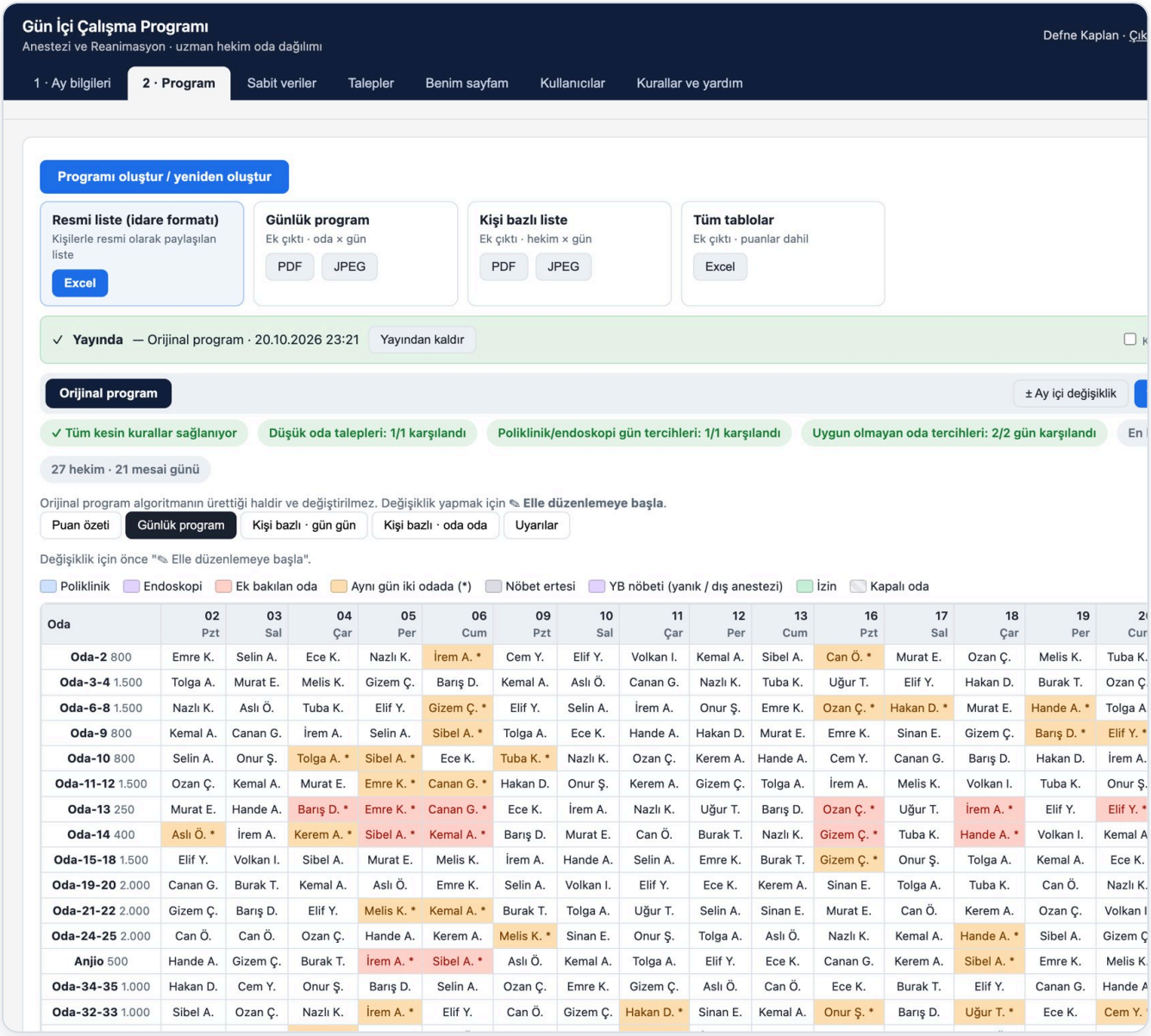
Each doctor's own page: request status and their personal monthly schedule.



The scheduler sees who has replied, a heat map of requested days, and approves or rejects each day with one click; approved days flow into the month's leave table.

THE SCHEDULE

A full month in seconds — and every hard rule kept



Oda	02 Pzt	03 Sal	04 Çar	05 Per	06 Cum	09 Pzt	10 Sal	11 Çar	12 Per	13 Cum	16 Pzt	17 Sal	18 Çar	19 Per	2(Cur
Oda-2 800	Emre K.	Selin A.	Ece K.	Nazlı K.	İrem A. *	Cem Y.	Elif Y.	Volkan I.	Kemal A.	Sibel A.	Can Ö. *	Murat E.	Ozan Ç.	Melis K.	Tuba K.
Oda-3-4 1.500	Tolga A.	Murat E.	Melis K.	Gizem Ç.	Barış D.	Kemal A.	Aslı Ö.	Canan G.	Nazlı K.	Tuba K.	Uğur T.	Elif Y.	Hakan D.	Burak T.	Ozan Ç.
Oda-6-8 1.500	Nazlı K.	Aslı Ö.	Tuba K.	Elif Y.	Gizem Ç. *	Elif Y.	Selin A.	İrem A.	Onur Ş.	Emre K.	Ozan Ç. *	Hakan D. *	Murat E.	Hande A. *	Tolga A
Oda-9 800	Kemal A.	Canan G.	İrem A.	Selin A.	Sibel A. *	Tolga A.	Ece K.	Hande A.	Hakan D.	Murat E.	Emre K.	Sinan E.	Gizem Ç.	Barış D. *	Elif Y. *
Oda-10 800	Selin A.	Onur Ş.	Tolga A. *	Sibel A. *	Ece K.	Tuba K. *	Nazlı K.	Ozan Ç.	Kerem A.	Hande A.	Cem Y.	Canan G.	Barış D.	Hakan D.	İrem A.
Oda-11-12 1.500	Ozan Ç.	Kemal A.	Murat E.	Emre K. *	Canan G. *	Hakan D.	Onur Ş.	Kerem A.	Gizem Ç.	Tolga A.	İrem A.	Melis K.	Volkan I.	Tuba K.	Onur Ş.
Oda-13 250	Murat E.	Hande A.	Barış D. *	Emre K. *	Canan G. *	Ece K.	İrem A.	Nazlı K.	Uğur T.	Barış D.	Ozan Ç. *	Uğur T.	İrem A. *	Elif Y.	Elif Y. *
Oda-14 400	Aslı Ö. *	İrem A.	Kerem A. *	Sibel A. *	Kemal A. *	Barış D.	Murat E.	Can Ö.	Burak T.	Nazlı K.	Gizem Ç. *	Tuba K.	Hande A. *	Volkan I.	Kemal A
Oda-15-18 1.500	Elif Y.	Volkan I.	Sibel A.	Murat E.	Melis K.	İrem A.	Hande A.	Selin A.	Emre K.	Burak T.	Gizem Ç. *	Onur Ş.	Tolga A.	Kemal A.	Ece K.
Oda-19-20 2.000	Canan G.	Burak T.	Kemal A.	Aslı Ö.	Emre K.	Selin A.	Volkan I.	Elif Y.	Ece K.	Kerem A.	Sinan E.	Tolga A.	Tuba K.	Can Ö.	Nazlı K.
Oda-21-22 2.000	Gizem Ç.	Barış D.	Elif Y.	Melis K. *	Kemal A. *	Burak T.	Tolga A.	Uğur T.	Selin A.	Sinan E.	Murat E.	Can Ö.	Kerem A.	Ozan Ç.	Volkan I
Oda-24-25 2.000	Can Ö.	Can Ö.	Ozan Ç.	Hande A.	Kerem A.	Melis K. *	Sinan E.	Onur Ş.	Tolga A.	Aslı Ö.	Nazlı K.	Kemal A.	Hande A. *	Sibel A.	Gizem Ç
Anjio 500	Hande A.	Gizem Ç.	Burak T.	İrem A. *	Sibel A. *	Aslı Ö.	Kemal A.	Tolga A.	Elif Y.	Ece K.	Canan G.	Kerem A.	Sibel A. *	Emre K.	Melis K.
Oda-34-35 1.000	Hakan D.	Cem Y.	Onur Ş.	Barış D.	Selin A.	Ozan Ç.	Emre K.	Gizem Ç.	Aslı Ö.	Can Ö.	Ece K.	Burak T.	Elif Y.	Canan G.	Hande A
Oda-32-33 1.000	Sibel A.	Ozan Ç.	Nazlı K.	İrem A. *	Elif Y.	Can Ö.	Gizem Ç.	Hakan D. *	Sinan E.	Kemal A.	Onur Ş. *	Barış D.	Uğur T. *	Ece K.	Cem Y.

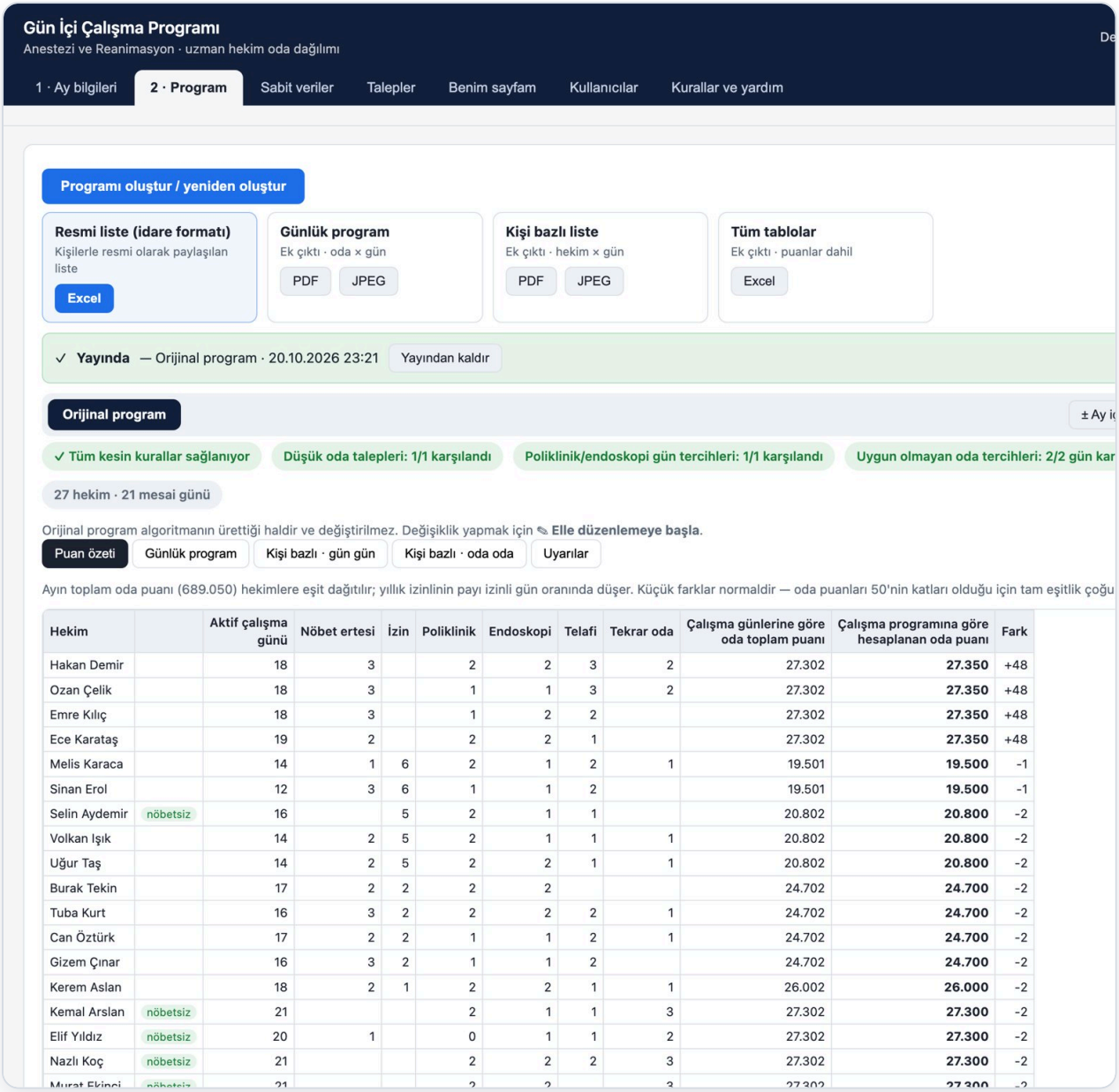
Room × day view of the generated month. Colours mark clinics and endoscopy; an asterisk marks a doctor covering two rooms that day. Status pills summarise rules and requests.

Under the hood: an optimisation algorithm (simulated annealing) explores millions of possible assignments.

- **15 hard rules** are never broken — e.g. no work the day after a night shift; a room with a location constraint can only be paired with specific rooms.
- **Preferences** are met where possible, in a fixed priority order.
- **Fairness**: monthly totals are pulled as close as possible to each doctor's fair target (reduced pro rata for leave).
- **Month to month**: doctors rotate through rooms they did not work last month, and clinic/endoscopy days alternate between 1 and 2.

FAIRNESS

Equal workload, measured



Hekim		Aktif çalışma günü	Nöbet ertesi	İzin	Poliklinik	Endoskopi	Telafi	Tekrar oda	Çalışma günlerine göre oda toplam puanı	Çalışma programına göre hesaplanan oda puanı	Fark
Hakan Demir		18	3		2	2	3	2	27.302	**27.350**	+48
Ozan Çelik		18	3		1	1	3	2	27.302	**27.350**	+48
Emre Kılıç		18	3		1	2	2		27.302	**27.350**	+48
Ece Karataş		19	2		2	2	1		27.302	**27.350**	+48
Melis Karaca		14	1	6	2	1	2	1	19.501	**19.500**	-1
Sinan Erol		12	3	6	1	1	2		19.501	**19.500**	-1
Selin Aydemir	nöbetsiz	16		5	2	1	1		20.802	**20.800**	-2
Volkan Işık		14	2	5	2	1	1	1	20.802	**20.800**	-2
Uğur Taş		14	2	5	2	2	1	1	20.802	**20.800**	-2
Burak Tekin		17	2	2	2	2			24.702	**24.700**	-2
Tuba Kurt		16	3	2	2	2	2	1	24.702	**24.700**	-2
Can Öztürk		17	2	2	1	1	2	1	24.702	**24.700**	-2
Gizem Çınar		16	3	2	1	1	2		24.702	**24.700**	-2
Kerem Aslan		18	2	1	2	2	1	1	26.002	**26.000**	-2
Kemal Arslan	nöbetsiz	21			2	1	1	3	27.302	**27.300**	-2
Elif Yıldız	nöbetsiz	20	1		0	1	1	2	27.302	**27.300**	-2
Nazlı Koç	nöbetsiz	21			2	2	2	3	27.302	**27.300**	-2

Point summary: each doctor's target (based on days actually worked) versus the points they received.

≈ 0.2%
largest gap between a doctor's total and their fair target in test runs on a real month (57 points out of ~27,000)

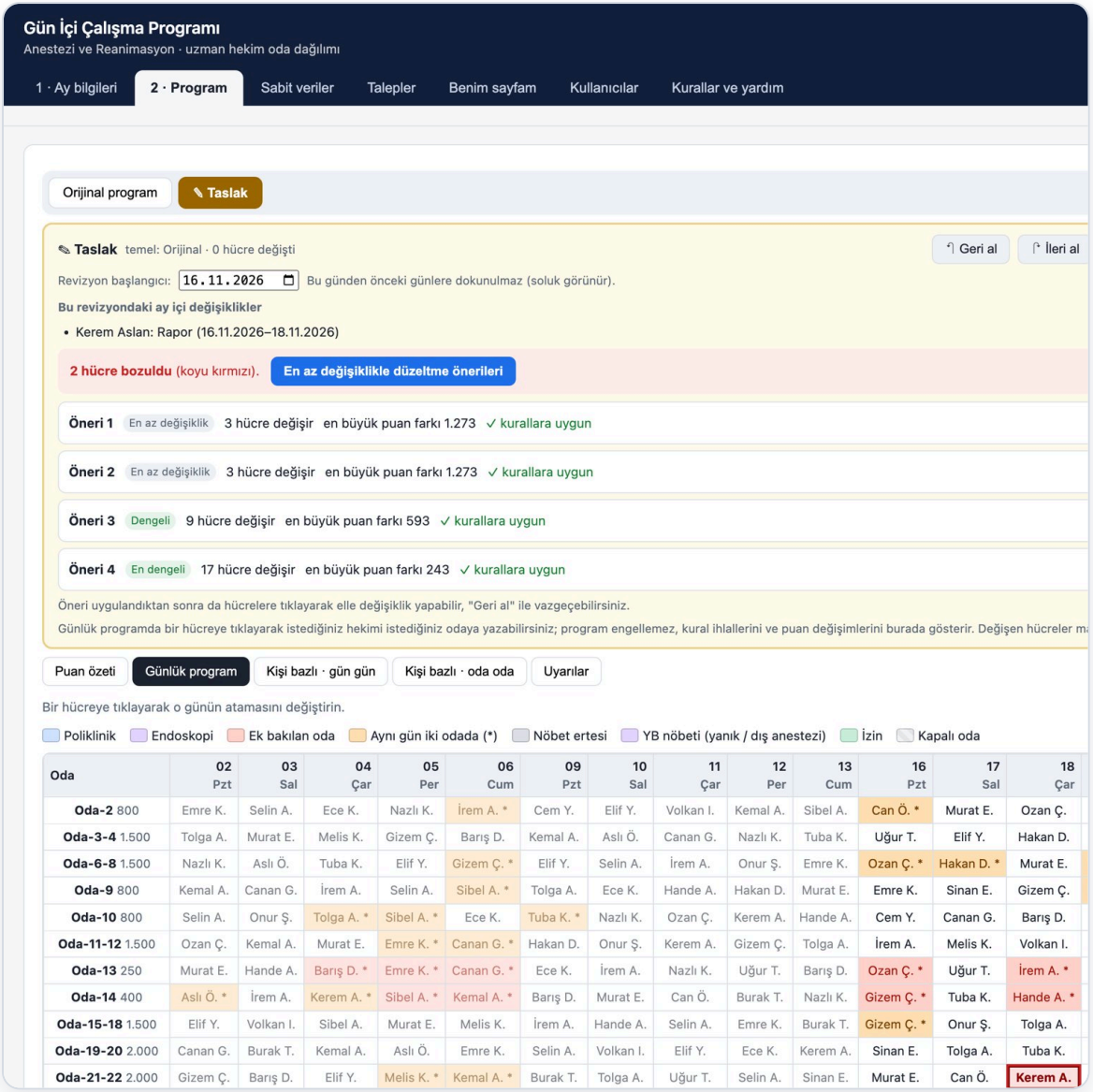
0
hard-rule violations

seconds
to compute a full month

Doctors on leave have their target reduced in proportion to the days they work, so leave never “costs” anyone points.

REAL LIFE HAPPENS

Mid-month changes without starting over



A doctor reports sick: affected cells turn dark red, past days stay locked, and the program proposes fixes — from “fewest changes” to “most balanced” — showing how many cells change and how fair each option is.

Çalışma Programı Değişiklik Bildirimi — Kasım 2026

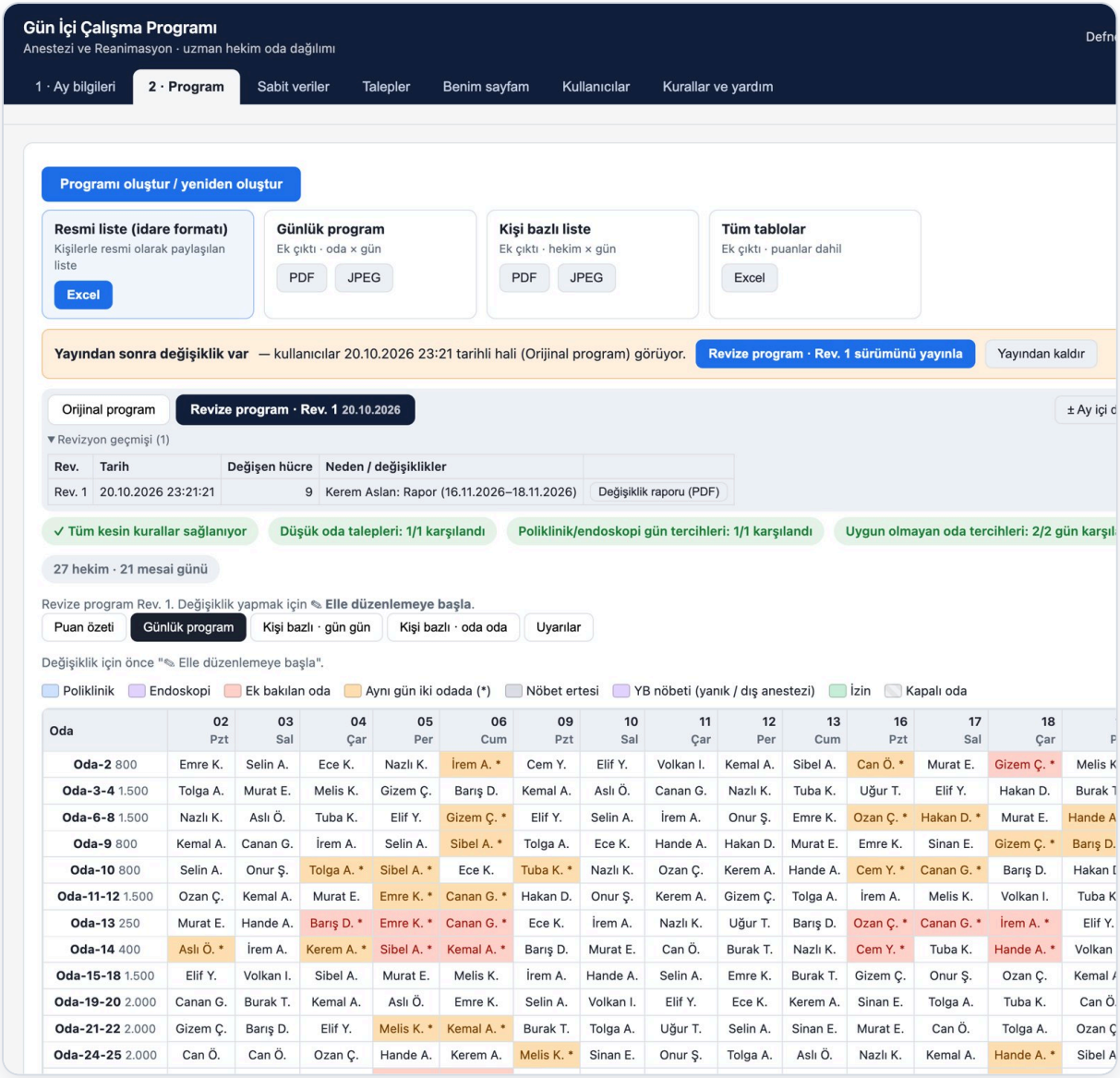
ŞEHİR EĞİTİM VE ARAŞTIRMA HASTANESİ ANESTEZİYOLOJİ VE REANİMASYON KLİNİĞİ · Revizyon 1 · 20.10.2026

Tarih	Gün	Oda / bölüm	Önceki	Yeni	Neden
16.11.2026	Pazartesi	Oda-14	Gizem Çınar (Oda-15-18 ile)	**Cem Yavuz (Oda-10 ile)**	yeniden düzenleme
17.11.2026	Salı	Oda-13	Uğur Taş	**Canan Güler (Oda-10 ile)**	yeniden düzenleme
17.11.2026	Salı	Anjio	Kerem Aslan	**Uğur Taş**	Kerem Aslan: Rapor (16.11.2026–18.11.2026)
18.11.2026	Çarşamba	Oda-2	Ozan Çelik	**Gizem Çınar (Oda-9 ile)**	yeniden düzenleme
18.11.2026	Çarşamba	Oda-15-18	Tolga Avcı	**Ozan Çelik**	yeniden düzenleme
18.11.2026	Çarşamba	Oda-21-22	Kerem Aslan	**Tolga Avcı**	Kerem Aslan: Rapor (16.11.2026–18.11.2026)
24.11.2026	Salı	Oda-28-30	Kerem Aslan	**İrem Aksoy**	yeniden düzenleme
24.11.2026	Salı	Septik	İrem Aksoy	**Burak Tekin**	yeniden düzenleme
24.11.2026	Salı	Kons-1	Burak Tekin	**Kerem Aslan**	yeniden düzenleme

Each finalised revision produces an official change report for the administration: date, room, previous doctor, new doctor, reason.

CONTROL STAYS WITH PEOPLE

Original, revised and draft versions



The algorithm's original schedule is never overwritten. Manual edits happen in a draft with undo/redo; the program never blocks the scheduler, but reports any rule broken and every change in points. Finalised drafts become numbered revisions with a full history.

Outputs

- Official list in the administration's Excel template
- Daily schedule and per-doctor schedule (PDF / image)
- Change report for every revision (PDF)
- Personal schedule for each doctor (PDF / Excel)

Privacy

- Doctors see only published schedules; access is by invitation only.
- The desktop version keeps its data in an encrypted file.

PROCESS · ROLE · STATUS

From a real need to a tool in use

Timeline (2026)

Oct 13	Department rules collected and written down; first algorithm
Oct 14	First working program
Oct 17–18	Personal requests, official administration format, step-by-step input
Oct 20	Colleagues tested it; their feedback implemented (room-pairing rule, closed rooms, manual edits, mid-month revisions)
Nov	Desktop version in use for the department's first real month
Next	Launch of the web version shown in this portfolio

Zeynep's role

- Identified the need and proposed the idea
- Helped turn the department's unwritten practice into precise rules and requirements
- Tested the program and gathered feedback from its users
- Took part in design and development sessions

How it was built

The program was built with AI-assisted development (Anthropic's Claude), through many design-and-test sessions driven by the department's real needs.

All names in this portfolio are fictional (personal data protection).